

# VISHWANATH M M

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## PROFESSIONAL SUMMARY

AI and Machine Learning undergraduate with hands-on experience in software engineering, backend development, systems programming, and open-source software. Founder and Lead Developer of the Ferrite programming language and maintainer of multiple Python packages published on PyPI. Proficient in Python, Rust, compiler construction, scalable software architecture, and backend system development. Seeking a Software Engineer or AI/ML Engineer role to build reliable, high-performance systems and developer tools.

## EDUCATION

|                                                                                 |                           |
|---------------------------------------------------------------------------------|---------------------------|
| <b>Bachelor of Engineering — Artificial Intelligence &amp; Machine Learning</b> | 2023 – Present            |
| Bapuji Institute of Engineering and Technology, Davangere                       | <b>CGPA: 7.93 / 10.0</b>  |
| <b>Class 12 (Pre-University)</b>                                                | 2023                      |
| Karnataka State Board                                                           | <b>Percentage: 85.16%</b> |
| <b>Class 10 (SSLC)</b>                                                          | 2021                      |
| Central Board of Secondary Education.                                           | <b>Percentage: 73.33%</b> |

## TECHNICAL SKILLS

**Languages:** Python, Rust, SQL.

**Backend & Frameworks:** REST API Design, Backend Development.

**Systems Programming:** Compiler Design, Systems Programming, Lexer/Parser Construction, CLI Tooling.

**Databases:** MongoDB, PostgreSQL.

**AI / ML:** Machine Learning, Natural Language Processing (NLP), LLM Integration.

**Frontend & Web:** HTML, CSS, React.

**DevOps & Tooling:** Git, GitHub, Docker, Linux, PyPI.

## EXPERIENCE

**Founder & Lead Developer — Ferrite Programming Language** 2025 – Present

**Website:** [ferrite-lang.org](https://ferrite-lang.org)

**GitHub:** [github.com/vishwanathdvgmm/ferrite](https://github.com/vishwanathdvgmm/ferrite)

- Founded Ferrite, an ML-oriented systems programming language written in Rust, comprising 3K+ lines of code, 10+ modules, and a 5-stage compiler pipeline.
- Created a modular compiler comprising 3K+ lines of Rust code across 10+ source modules, covering lexical analysis, parsing, AST generation, semantic analysis, and execution.
- Architected a scalable compiler architecture spanning 10+ Rust modules with 30+ unit test files, supporting extensibility, memory safety, and structured diagnostics.
- Author and maintain 30+ unit test files to validate compiler components and improve development reliability.
- Manage the project roadmap, documentation, GitHub repository, website, release planning, and semantic versioning across 20+ commits.

## PROJECTS

**Open-Source Python Packages (PyPI)** 2024 – Present

**Open-Source Project**

**Tech Stack:** Python, PyPI, PEP 508, Package Development, Dependency Resolution.

- Published a production-ready Python package that automates the consolidation of multiple virtual environments, reducing environment setup time by over **90%** for multi-project workflows.
- Introduced **PEP 508-compliant dependency analysis** and conflict detection, enabling safe merging of Python environments while identifying version incompatibilities before execution.
- Released an open-source Python library providing a modular API for Unicode symbols, emojis, arrows, mathematical operators, currency symbols, and other special characters for Python applications.
- Maintained both packages with modular architectures, semantic versioning, automated testing, and comprehensive documentation for PyPI distribution.
- **GitHub:** [github.com/vishwanathdvgmm/pyvenvmerge](https://github.com/vishwanathdvgmm/pyvenvmerge) | [github.com/vishwanathdvgmm/symjoy](https://github.com/vishwanathdvgmm/symjoy)

**Tech Stack:** Python, Go, MongoDB, React, Blockchain

- Developed a blockchain-integrated digital evaluation platform to provide secure, transparent, and tamper-evident management of academic assessment records.
- Engineered a Python backend supporting end-to-end workflows for exam submission, automated evaluation, and blockchain-based record verification.
- **Leveraged** MongoDB as the off-chain datastore and incorporated SHA-256 hash generation with blockchain anchoring to ensure data integrity and detect record tampering.
- Built a React-based web interface enabling administrators and students to submit, manage, and verify evaluation records through a centralized dashboard.
- **GitHub:** [github.com/vishwanathdvghmm/digital-eval-system](https://github.com/vishwanathdvghmm/digital-eval-system)

CERTIFICATIONS & TRAINING

- |   |                                                                        |      |
|---|------------------------------------------------------------------------|------|
| • | <b>Hashgraph Developer Course</b> — The Hashgraph Association          | 2025 |
| • | <b>Building AI-Powered Search with MongoDB Vector Search</b> — MongoDB | 2026 |
| • | <b>Artificial Intelligence Training</b> — Acmegrade                    | 2026 |